Joint Routing and Wavelength Assignment with Deep Reinforcement Learning for Quantum-Classical Coexistence in WDM Optical Networks

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***Abstract*—This paper studies joint routing and wavelength assignment (RWA) for quantum key distribution (QKD) coexisting with classical traffic in WDM optical networks. Four-wave mixing (FWM) and spontaneous Raman scattering (SpRS) from classical channels degrade per-link key capacity in a wavelength-dependent manner, creating a dual-constrained resource allocation problem. We propose a two-layer framework: at the routing layer, four path-selection strategies with increasing QKD awareness, culminating in a dual-capacity-aware (DCA) strategy that balances classical-bandwidth and QKD-key residuals when ranking candidate paths; and at the wavelength assignment layer, a DQN agent that learns to place classical channels so as to preserve per-link key capacity. Simulation on the NSFNET 14-node topology shows that, given identical $k = 3$ candidate paths, the DQN wavelength agent reduces the request blocking rate by 8.5–10.4% and improves the aggregate secret key rate by 4–5% over first-fit assignment across all four routing strategies, while the distance- and dual-capacity-aware routing strategies further reduce blocking by favoring shorter, lower-noise paths.**

***Index Terms*—Quantum key distribution, routing and wavelength assignment, deep reinforcement learning, four-wave mixing, Raman scattering, WDM optical networks**

I. Introduction

Integrating QKD into existing WDM optical networks is economically attractive, as it reuses deployed fiber infrastructure instead of requiring dedicated dark fibers [2]. However, the extreme power disparity between classical (0–10 dBm) and quantum (single-photon-level) signals causes two dominant nonlinear impairments: four-wave mixing (FWM) and spontaneous Raman scattering (SpRS) [6], [7]. Both effects depend on *which wavelengths* carry classical traffic, so the secret key rate (SKR) available on each link varies with the current wavelength occupancy.

This creates a **dual-constrained RWA problem**: each link provides classical bandwidth (Gbps) and key capacity (kbps), where the latter changes dynamically with traffic state. Prior work either assumes fixed key capacities [3] or proposes static wavelength plans without dynamic traffic [4]. Deep reinforcement learning (DRL) has shown promise for classical RWA [5] but has not been applied to QKD-aware coexistence networks.

This paper makes three contributions: (1) a QKD-aware RWA problem formulation with state-dependent key capacities derived from discrete FWM and SpRS noise models; (2) four routing strategies with increasing dual-capacity awareness evaluated on the NSFNET topology under Poisson traffic; and (3) a DQN wavelength assignment agent trained on a custom Gymnasium environment that outperforms first-fit on blocking rate and aggregate SKR.

II. Network Modeling

A. WDM Network Topology and Traffic Model

We model the optical network as a directed graph $G = (V, E)$, where vertices represent optical switching nodes and edges represent fiber spans. We use the NSFNET 14-node, 21-edge topology as the reference network, with edge lengths ranging from 52.5 km to 300 km. Each edge carries W WDM wavelengths on the ITU-T G.694.1 C-band grid [11], with center frequencies

$$f(w) = f_{\text{start}} + (w-1)\cdot \Delta f_{\text{ch}}, \quad w \in \{1,\dots,W\}, \tag{1}$$

where $f_{\text{start}} = 190.1$ THz, $\Delta f_{\text{ch}} = 50$ GHz (50-GHz fixed grid, C-band), and $W = 32$ slots are available for classical traffic in our simulation. Each 50-GHz slot carries one DP-16QAM lightpath at 100 Gbps.

Service requests arrive following a Poisson process with rate λ and hold for exponentially distributed durations with mean $1/\mu$, yielding offered load $\rho = \lambda/\mu$ Erlangs. Each request r specifies a source-destination pair (s_r, d_r), a classical bandwidth demand b_r (Gbps), and a key consumption demand k_r (kbps). Because each slot carries 100 Gbps, a request with demand b_r requires

$$n_r = \lceil b_r/100 \rceil \tag{2}$$

wavelength slots on every link of its path.

B. Dual-Capacity Resource Model

Each edge $e \in E$ provides two types of capacit

- **Classical capacity** C_e^{cl} (Gbps): determined by the modulation format and spectral efficiency of active classical wavelengths.
- **Key capacity** $K_e(\mathbf{x}_e)$ (kbps): the aggregate secret key rate supported by quantum channels on edge e, which depends on the binary occupancy vector $\mathbf{x}_e \in \{0,1\}^W$ indicating which wavelengths carry classical traffic on that edge.

The key capacity is computed from the SKR model after evaluating the FWM and SpRS noise generated by all active classical wavelengths. When the classical occupancy of a link changes (due to an arrival or departure), the key capacity is recalculated.

Limitation: This model does not evaluate the classical channel signal-to-noise ratio (SNR) against the DP-16QAM threshold. In a real deployment, the nonlinear interference (NLI) from FWM and cross-phase modulation must be verified to remain below the required OSNR ($\approx$ 18 dB) for 100-Gbps DP-16QAM transmission. Incorporating a per-channel NLI penalty model (e.g., GNPy) is left for future work.

C. Discrete Single-Frequency Noise Models

We adopt the discrete single-frequency approach, treating each classical channel as a monochromatic pump at its ITU grid center frequency with launch power P_{ch}.

1) FWM Noise: For a quantum channel at frequency f_q, the forward FWM noise power at fiber length L is computed by enumerating all classical channel triplets (i,j,k) satisfying the frequency-matching condition $f_i + f_j - f_k = f_q$ [6]:

$$P_{\text{FWM}}^{\text{fwd}}(f_q, L) = \frac{\gamma^2 e^{-\alpha_q L}}{9} \sum_{\substack{(i,j,k) \\ f_i+f_j-f_k=f_q}} D_{ijk}^2 \cdot \eta_{ijk}(L) \cdot P_i P_j P_k, \tag{3}$$

where γ is the fiber nonlinear coefficient, $D_{ijk} \in \{3,6\}$ is the degeneracy factor, and $\eta_{ijk}(L)$ is the FWM efficiency:

$$\eta_{ijk}(L) = \left| \frac{1 - \exp[-\alpha L + j\Delta\beta_{ijk} L]}{\alpha - j\Delta\beta_{ijk}} \right|^2, \tag{4}$$

with phase mismatch expanded to second-order dispersion:

$$\begin{aligned}\Delta\beta_{ijk} \approx \frac{2\pi\lambda^2}{c}(f_i - f_k)(f_j - f_k)\Big[D_c \\ + \frac{\lambda^2}{2c}\frac{dD_c}{d\lambda}(f_i + f_j - 2f_0)\Big].\end{aligned} \tag{5}$$

Backward FWM arises from Rayleigh backscattering of the forward-generated FWM power integrated along the fiber span. The total FWM noise power is $P_{\text{FWM}} = P_{\text{FWM}}^{\text{fwd}} + P_{\text{FWM}}^{\text{bwd}}$.

2) SpRS Noise: The SpRS noise from a classical pump at f_p to a quantum channel at f_q is computed via the Raman cross-section with phonon population correction [7]:

$$P_{\text{SpRS}} = \sum_{p \in \mathcal{C}} P_p \cdot \sigma(f_p, f_q) \cdot [n_{\text{th}}(\Delta f) + \Theta(\Delta f)] \cdot \mathcal{P}(f_p, f_q, L), \tag{6}$$

where $\Delta f = f_p - f_q$, $n_{\text{th}}(\Delta f) = [\exp(h|\Delta f|/kT) - 1]^{-1}$ is the phonon occupation factor, $\Theta(\Delta f) = 1$ for Stokes ($\Delta f > 0$) and 0 otherwise, $\sigma(f_p, f_q)$ incorporates the measured 92-point GNPy Raman gain coefficient $g_R(|\Delta f|)$, and $\mathcal{P}$ is the distributed propagation factor distinguishing forward and backward scattering.

D. Secret Key Rate Computation

The noise photon probability per detector gate is

$$p_{\text{noise}} = \frac{P_{\text{FWM}} + P_{\text{SpRS}}}{2hf_q} \cdot \tau_{\text{SPD}} \cdot \eta_{\text{SPD}} \cdot \text{IL}_{\text{post}}. \tag{7}$$

The SKR under the three-intensity decoy-state BB84 protocol [1] is lower-bounded by [8]:

$$R \geq p_\mu \, q \left\{ -Q_\mu f_e H_2(E_\mu) + Q_1^L \left[1 - H_2(e_1^U)\right] \right\}, \tag{8}$$

where p_μ is the probability of sending a signal state, q depends on the QKD protocol ($q = 0.5$ for decoy-state BB84), f_e is the error-correction efficiency, $H_2(\cdot)$ is the binary Shannon entropy, Q_1^L and e_1^U are respectively the lower bound on the single-photon gain and the upper bound on the single-photon error rate, and Q_μ and E_μ are the signal-state gain and error rate. Finite-key effects ($N = 10^{10}$ pulses) are incorporated following [9], Eqs. (26)–(36). Protocol parameters are $\mu = 0.75$ (signal), $\nu = 0.2$ (decoy), $f_e = 1.16$, detector efficiency $\eta_{\text{SPD}} = 0.25$, dark count probability $p_{\text{dark}} = 10^{-6}$, and gate width $\tau_{\text{SPD}} = 1$ ns. The insertion loss $\text{IL} = 6$ dB is applied after the shared fiber span.

A link e is *reliable* if $R > 0$ for all quantum channels on that link given the current classical occupancy $\mathbf{x}_e$. The total key capacity of link e is $K_e(\mathbf{x}_e) = N_q \cdot R \cdot R_{\text{rep}}$, where N_q is the number of parallel quantum channels and $R_{\text{rep}} = 50$ MHz is the pulse repetition rate.

Figure 1 shows the finite-key SKR vs. fiber distance for the default protocol parameters, computed using the self-contained `qkd_routing/physics.py` module. On a dark fiber ($p_{\text{noise}} = 0$), the SKR remains positive out to roughly 150 km. With realistic SpRS noise from adjacent classical channels the achievable distance shortens sharply—to about 80 km with 2 co-propagating channels and below 50 km with 8—motivating the joint routing and wavelength assignment optimization.

III. Algorithm Design

A. Problem Formulation

Each request r requires $n_r = \lceil b_r/100 \rceil$ wavelength slots (from (2)). The joint routing and wavelength assignment (RWA) problem is:

$$\begin{aligned} \underset{p,\, \mathcal{W}_r}{\text{maximize}} \quad & U(p, \mathcal{W}_r; \mathbf{X}) \\ \text{subject to} \quad & |\mathcal{W}_r| = n_r, \\ & w \notin \mathbf{x}_e,\ \forall w \in \mathcal{W}_r,\ \forall e \in p, \\ & K_e(\mathbf{x}_e \cup \mathcal{W}_r) \geq k_r,\ \forall e \in p, \end{aligned} \tag{9}$$

where $\mathbf{X}$ is the global wavelength occupancy state, $\mathcal{W}_r \subseteq \{1, \ldots, W\}$ is the set of slots assigned to request r, and

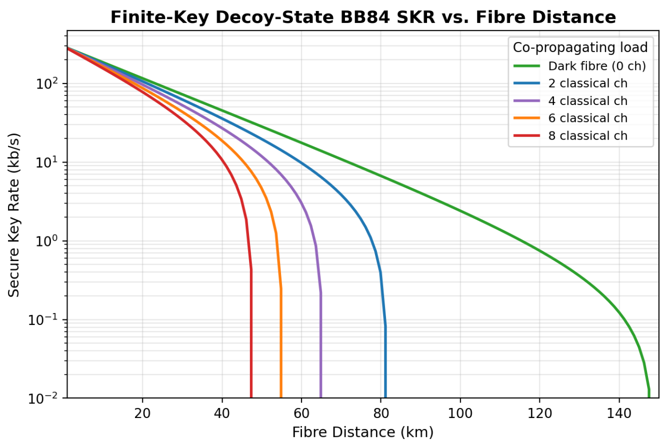

Fig. 1. Finite-key decoy-state BB84 SKR vs. fiber distance for 0–8 co-propagating classical channels at -25 dBm/channel.

$U(\cdot)$ is a network utility function. The second constraint is the *wavelength continuity constraint*: each assigned slot must be free on *every* link of the chosen path (since we assume all-optical switching with no wavelength conversion). The third constraint ensures the post-assignment key rate meets the demand k_r.

We decompose the problem into two stages: (1) *routing* selects a candidate end-to-end path, and (2) *wavelength assignment* selects a specific wavelength on that path.

B. Routing Strategies

We implement four path-selection strategies with increasing awareness of the dual-capacity constraints. All strategies operate over the $K = 5$ shortest paths precomputed for each node pair.

- **Min-Hop (MH)**: Select the path with the fewest hops; ties broken by shortest distance. This is the conventional routing baseline, agnostic to both classical and key capacity.
- **Min-Distance (MD)**: Select the shortest-distance path; ties broken by fewest hops. Slightly more key-capacity-aware than MH (shorter fiber $\to$ lower attenuation $\to$ higher SKR), but still capacity-oblivious.
- **Key-Capacity-Aware (KCA)**: Among all feasible paths (wavelength slots available and key demand met), select the one maximizing the bottleneck key residual ratio:

$$\max_{p} \min_{e \in p} \frac{K_e^{\text{residual}}}{k_r}, \tag{10}$$

  where K_e^{residual} is the remaining key capacity on edge e after accounting for currently active connections. Ties are broken by shortest distance.
- **Dual-Capacity-Aware (DCA)**: Jointly consider both resource dimensions, maximizing the normalized dual-resource bottleneck:

$$\max_{p} \min_{e \in p} \min\left(\frac{C_e^{\text{residual}}}{b_r}, \frac{K_e^{\text{residual}}}{k_r}\right). \tag{11}$$

  This strategy balances classical bandwidth and key capacity, avoiding paths where either resource is scarce even if the other is abundant.

C. Wavelength Assignment

Given a path selected by the routing layer, a wavelength must be assigned from the set of feasible slots—those free on all edges of the path and not violating the SKR reliability constraint. We compare two policies.

1) First-Fit (Benchmark): The first-fit (FF) policy scans wavelengths in increasing index order and assigns the first feasible slot. FF is computationally efficient ($O(W \cdot |p|)$) and serves as the baseline for all four routing strategies.

2) DQN-Based Wavelength Selection: First-fit is resource-oblivious: it does not consider how the wavelength choice affects key capacity for future requests. We train a deep Q-network (DQN) agent that selects wavelengths to minimize the long-term impact on network key capacity.

a) Environment Formulation: We model wavelength assignment as a Markov decision process (MDP). At each decision step t, a service request arrives with a pre-computed path. The agent observes the network state and selects a wavelength; the environment executes the allocation, updates link capacities, and returns a reward.

- **State** s_t: A vector encoding (i) the binary wavelength occupancy matrix $\mathbf{X} \in \{0,1\}^{|E| \times W}$ showing which wavelengths are occupied on which edges, (ii) the requested path p as an edge-index mask, (iii) the quantum channel positions on each edge, and (iv) the request parameters (b_r, k_r).
- **Action** $a_t \in \{1, \ldots, W\}$: The wavelength index to assign. An action mask eliminates infeasible wavelengths (occupied on any edge of p, or would violate the SKR reliability constraint).
- **Reward** r_t: A composite accept/preserve reward,

$$r_t = \begin{cases} 1 + 2\,\min(\rho_{\text{SKR}}, 1), & \text{if accepted,} \\ -3, & \text{if blocked,} \end{cases} \tag{12}$$

  where $\rho_{\text{SKR}} = \min_{e \in p} \text{SKR}_e^{\text{after}} / \text{SKR}_e^{\text{before}}$ is the bottleneck post-assignment key ratio across the path edges, clipped to $[0, 1]$. This yields $r_t \in [+1, +3]$ for accepted requests, so the agent is rewarded for accepting while preserving key capacity (placements that drive heavier FWM/SpRS noise onto the quantum channel lower ρ_{SKR} and hence the reward), and penalised for blocking. The break-even acceptance rate $3/5 = 60\%$ lies well below the observed rate, giving a positive learning signal.

b) Network Architecture and Training: We implement the DQN agent [10] with a dueling Q-network: a shared feature extractor (three fully-connected layers of 1024, 1024, and 512 units with ReLU activations) followed by separate value and advantage streams. Action masking is applied at inference by setting infeasible actions' Q-values to $-\infty$ before argmax selection. Key hyperparameters: ϵ-greedy exploration decaying from 0.9 to 0.05 over 5×10^5 steps; replay buffer

10^6, batch size 4096, learning starts after 10,000 steps; target network updated every 5,000 steps; Adam optimizer, learning rate 3×10^{-4}, Huber loss. The best checkpoint (step ≈110K, eval reward 852) is selected by an evaluation callback that runs 10 masked episodes every 5K steps; training early-stops after 20 evaluations without improvement (here at ≈210K steps). The agent is trained on real K-shortest candidate paths sampled between random node pairs, so it learns wavelength assignment on multi-edge lightpaths rather than isolated links.

The environment is a custom Gymnasium environment wrapping the discrete-event simulation engine, the FWM/SpRS noise solvers, and the SKR computation, running on 16 parallel SubprocVecEnv instances.

D. Complexity Analysis

The per-request time complexity is $O(K \cdot |V|)$ for KSP-based routing, $O(W \cdot |p|)$ for FF wavelength assignment, and $O(W)$ for DQN inference (a single forward pass). The offline training cost is $O(T \cdot |E| \cdot W^3)$ due to FWM triplet enumeration during environment simulation, with $T \lesssim 5 \times 10^5$ steps. In deployment, only the lightweight inference pass is needed.

IV. Simulation Analysis

A. Simulation Setup

Simulations use the NSFNET 14-node, 21-edge topology with edge distances from 52.5 to 300 km. Fiber parameters follow standard SMF (ITU-T G.652): attenuation $\alpha = 0.2$ dB/km, $\gamma = 1.3$ W^{-1}km^{-1}, $D_c = 17$ ps/(nm·km), $dD_c/d\lambda = 0.056$ ps/(nm^2·km). The WDM grid uses 50 GHz channel spacing with $W = 32$ classical wavelengths (190.0–191.55 THz) and one quantum channel centered at 190.75 THz (≈slot 15). Classical launch power is -25 dBm per channel. Key capacity is computed using the embedded finite-key decoy-state BB84 model (Eq. 8) with the discrete FWM/SpRS noise solver from `qkd_routing/physics.py`.

Traffic is generated via a Poisson process with offered load $\rho \in [50, 400]$ Erlangs, mean holding time $1/\mu = 1$ s. Three security levels are mixed: 50% low-security (10–40 Gbps, 0.2 kbps key), 30% medium (40–100 Gbps, 1.0 kbps key), and 20% high-security (100–400 Gbps, 2.0 kbps key). Each edge supports $W = 32$ classical DWDM channels at 100 Gbps each (3200 Gbps per edge). The simulation runs 5000 requests per load point with 20% warmup.

B. Routing Strategy Performance

Fig. 2 shows the total blocking rate across 50–400 Erlangs; classical-bandwidth blocking is essentially zero in this regime, so all blocking is due to QKD key capacity. Fig. 3 compares the four strategies' key-blocking rate relative to min-hop: the strategies differ by at most ≈ 1.2 percentage points, with min-distance generally lowest at low-to-medium load. Fig. 4 shows the average classical utilisation, which stays below 20% across all loads.

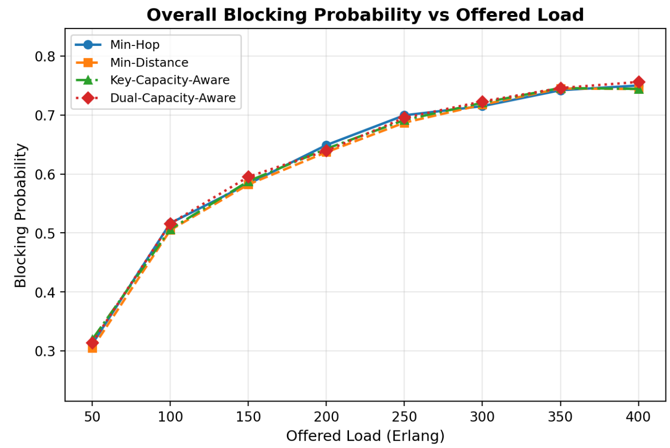

Fig. 2. Total blocking rate vs. offered load for the four routing strategies.

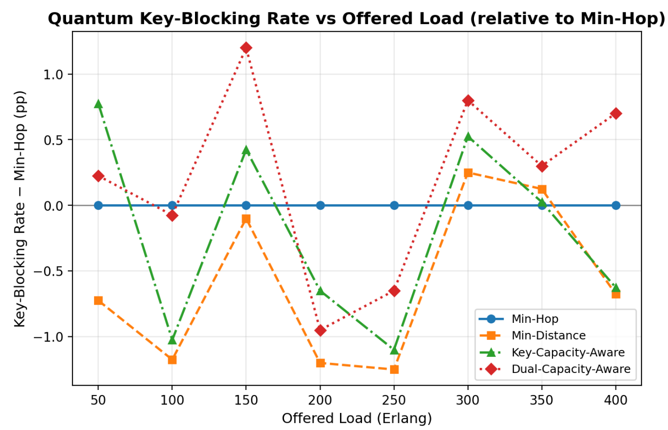

Fig. 3. Quantum key-blocking rate relative to Min-Hop vs. offered load for the four routing strategies.

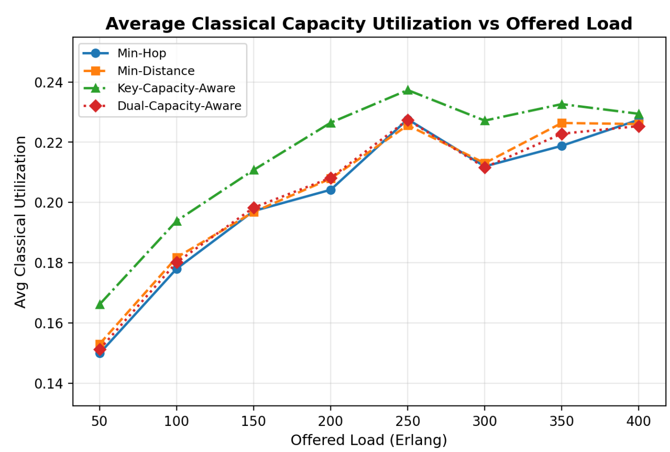

Fig. 4. Average classical-channel utilisation vs. offered load for the four routing strategies.

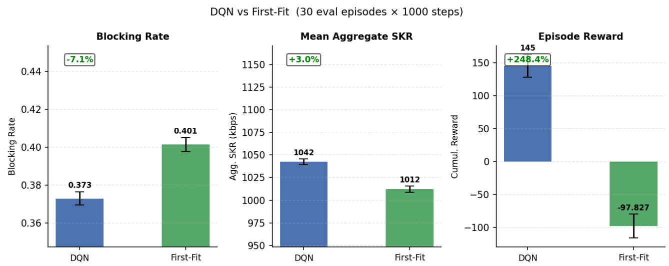

Fig. 5. DQN vs. first-fit wavelength assignment (30 evaluation episodes, 1000 steps each): blocking rate, aggregate SKR, and episode reward.

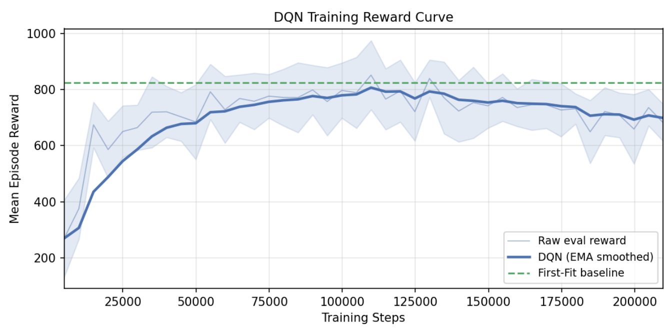

Fig. 6. DQN training reward curve (10 masked evaluation episodes every 5K steps), with the first-fit baseline shown for reference.

C. *DQN Wavelength Selection*

The quantum channel is placed at 190.75 THz (slot 15 in the 32-slot, 50 GHz-spaced grid), creating an asymmetric interference profile: wavelengths near slot 15 incur heavier Raman scattering onto quantum receivers than those at the grid extremes. The DQN learns to exploit this asymmetry by preferring slots distant from the quantum channel.

Figs. 5 and 6 compare DQN against first-fit (FF) over 30 evaluation episodes of 1000 steps each. Key results are summarised in Table I.

TABLE I
DQN VS. FIRST-FIT (30 EPISODES × 1000 STEPS, SEED 100)

Metric	DQN	First-Fit
Blocking Rate	**37.30%**	40.13%
Classical Util.	**3.34%**	3.20%
Agg. SKR (kbps)	**1042.5**	1012.1
Episode Reward	**145.2**	−97.8

Here each request is presented with a single candidate path, isolating the effect of wavelength assignment. The DQN achieves a -7.1% reduction in blocking rate relative to first-fit (37.30% vs. 40.13%) and a $+3.0\%$ gain in aggregate SKR (1,043 kbps vs. 1,012 kbps); the higher absolute blocking than the joint setting below reflects the absence of a multi-path routing fallback. Blocking reason analysis shows that mixed classical-and-key blocks account for $\approx 60\%$ of all blocked requests in both policies, indicating that the bottleneck lies in the joint resource constraint rather than pure wavelength availability. The DQN's improvement stems from learning to avoid FWM-resonant wavelength placements that would degrade the quantum channel SKR, effectively translating physical-layer noise awareness into better long-term resource decisions.

D. *Joint Routing and Wavelength Assignment*

The experiment above isolates wavelength assignment on a fixed path. We now integrate the two layers: for each arriving request a routing strategy ranks the K shortest candidate paths, and wavelength assignment is attempted on each path *in order* until one succeeds (first-feasible wins, $k = 3$ candidates), otherwise the request is blocked. We compare the DQN and first-fit wavelength-assignment policies on *identical* candidate paths under all four routing strategies; results (10 episodes × 1000 steps, seed 100) appear in Table II and Fig. 7.

TABLE II
JOINT ROUTING + WA: DQN VS. FIRST-FIT WAVELENGTH ASSIGNMENT ($k = 3$ CANDIDATE PATHS, 10 EPISODES × 1000 STEPS)

	Blocking		Agg. SKR (kbps)	
Routing	**FF**	**DQN**	**FF**	**DQN**
Dual-Capacity-Aware	13.63%	**12.47%**	1019	**1064**
Min-Distance	13.12%	**11.87%**	1021	**1060**
Min-Hop	13.14%	**11.78%**	1021	**1059**
Key-Capacity-Aware	10.90%	**9.88%**	928	**972**

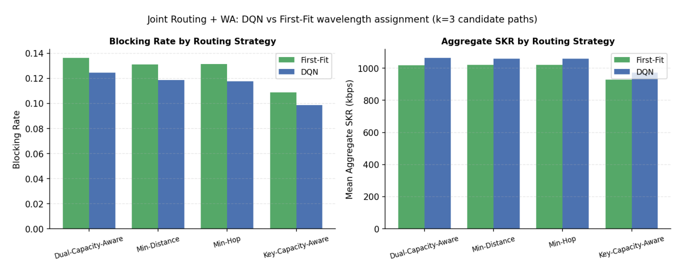

Fig. 7. Joint routing + wavelength assignment: blocking rate (left) and aggregate SKR (right) for first-fit vs. DQN wavelength assignment under each routing strategy.

Across all four routing strategies the DQN wavelength agent consistently lowers blocking (by 8.5–10.4% relative to first-fit) and raises aggregate SKR (by 4.0–4.7%), confirming that the noise-aware wavelength placement learned during training transfers to the multi-path routing setting. The gain is retained under the dual-capacity-aware (DCA) routing strategy, so the two layers are complementary: DCA steers requests onto key-rich, low-noise paths while the DQN packs classical channels to minimise Raman crosstalk onto the quantum band.

V. CONCLUSION

This paper has formulated the joint routing and wavelength assignment problem for QKD-classical coexistence in WDM optical networks and proposed a two-layer solution framework integrating capacity-aware routing with deep reinforcement learning for wavelength selection.

On the NSFNET 14-node topology, the dual-capacity-aware (DCA) routing strategy combined with the DQN wavelength agent reduces request blocking by 8.5% and raises the aggregate secret key rate by 4.3% relative to first-fit wavelength assignment on identical $k = 3$ candidate paths, and the DQN's advantage holds consistently (8.5–10.4% lower blocking) across all four routing strategies.

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APPENDIX: REPRODUCING RESULTS

All results can be reproduced from the project root with the `qkd_env` conda environment.

```
# 1. Routing simulation (4 strategies, load 50-400 Erlang):
python run.py --qkd-capacity-mode actual_skr \
  --load-start 50 --load-end 400 --load-step 50 \
  --num-requests 5000 --seed 42

# 2. DQN training (real k-shortest paths, GPU,
#    500K budget, early-stopped ~140K steps):
python -m src.rl.train_dqn --total-timesteps 500000 \
  --learning-rate 3e-4 --learning-starts 10000 \
  --target-update-interval 5000 --eval-freq 5000 \
  --exploration-final-eps 0.05 --n-eval-episodes 10 \
  --max-steps 2000 --device cuda

# 3. Wavelength-assignment eval (DQN vs First-Fit):
python -m src.rl.eval_rl --n-episodes 30 \
  --max-steps 1000 --seed 100

# 4. Joint routing + WA (k=3 candidate paths):
python -m src.rl.eval_joint_rwa --n-episodes 15 \
  --max-steps 1000 --seed 100
```